



Graphs in Machine Learning

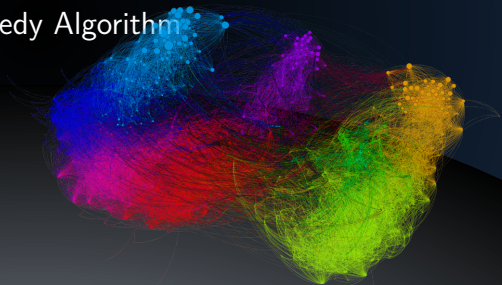
Submodularity: Theory

Definition, Properties, and Greedy Algorithm

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Partially based on material by: Andreas Krause,
Branislav Kveton, Michael Kearns



Submodularity: modeling diminishing returns

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A **set function** on a discrete set A is **submodular** if for any $S \subseteq T \subseteq A$ and for any $e \in A \setminus T$

$$f(S \cup \{e\}) - f(S) \geq f(T \cup \{e\}) - f(T)$$

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Link to our **product placement** problem on a **social network graph**?

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Let $S^* = \arg \max_{S \subseteq A, |S| \leq k} f(S)$ where f is monotonic and submodular set function and let S_{Greedy} be a **greedy solution**.

$$\text{Then } f(S_{\text{Greedy}}) \geq \left(1 - \frac{1}{e}\right) \cdot f(S^*).$$

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Other applications: information, graph cuts, covering, ...

Submodularity: Greedy algorithm

```
1: Input:  
2:    $k$ : the maximum allowed cardinality of the output  
3:    $V$ : a ground set  
4:    $f$ : a monotone, non-negative, and submodular function  
5: Run:  
6:  $S_0 = \emptyset$   
7: for  $i = 1$  to  $k$  do  
8:    $S_i \leftarrow S_{i-1} \cup \left\{ \arg \max_{a \in V \setminus S_{i-1}} [f(\{a\} \cup S_{i-1}) - f(S_{i-1})] \right\}$   
9: end for  
10: Output:  
11:   Return  $S_{\text{Greedy}} = S_k$ 
```

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Submodularity: Approximation guarantee of Greedy

Let S_i be the i -th set selected by Greedy. We show

$$f(S^*) - f(S_{i-1}) \leq f(S^* \cup S_{i-1}) - f(S_{i-1})$$

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Difference from the optimum of $S_{\text{Greedy}} = S_k$ after the k -th step

...

$$\begin{aligned} f(S^*) - f(S_k) &= f(S^*) - f(S_{k-1}) - (f(S_k) - f(S_{k-1})) \\ &\leq f(S^*) - f(S_{k-1}) - \frac{f(S^*) - f(S_{k-1})}{k} \\ &\leq \left(1 - \frac{1}{k}\right) \cdot (f(S^*) - f(S_{k-1})) \leq \left(1 - \frac{1}{k}\right)^k \cdot f(S^*) \end{aligned}$$

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back to the influence-maximization example ...

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<https://misovalko.github.io/mva-ml-graphs.html>